

UniCompass: Helping High School Students Find the Right College Major

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ABSTRACT

With the objective of furthering the United Nations' mission to provide Quality Education, this project aims to aid high school students in Taiwan's high-pressure, exam-oriented educational environment, by offering support and resources to help them uncover their interests and potential career paths. Through in-depth interviews with high school students, we identified common concerns in knowing and exploring college majors and career paths due to a lack of information and resources, unvarying suggestions, and barriers during the process of preparing for Academic Portfolio. To address these issues, we designed UniCompass using a user-centered design process. UniCompass is featured by: 1) finding the appropriate major for the student and comparing the differences between various universities and departments; 2) obtaining valuable information from the personal experiences shared by university students; and 3) receiving advice and guidance on building Academic Portfolios.

CCS CONCEPTS

• **Human-centered computing** → **User centered design.**

KEYWORDS

exam-oriented education; Academic Portfolio; major and career exploration website; university departments; information-providing

ACM Reference Format:

LIN, LIANG-CHENG, LAI, YI-CHUN, CHANG, WEI-CHIEN, CHIU, HSIN-LUN, and CHEN, TZU-YU. 2023. UniCompass: Helping High School Students Find the Right College Major. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3583848>

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3583848>

1 INTRODUCTION

In Taiwan, the prevalent culture and societal pressures drive students to follow a prescribed path for success, leading high school students to prioritize majors with high earning potential and high social status over personal interests and passions[5]. Additionally, the emphasis on examination scores in Taiwan's college admission process has led many high school students to turn to cram schools, a popular form of after-school tutoring in Taiwan, to boost their exam and academic performance[3]. Cram school classes consume a significant amount of time for students, which also limits their ability to explore majors that align with their personal interests[6].

Recently, in 2019, Taiwan proposed the Academic Portfolio System (APS) as a way allowing students to record and showcase their learning experiences and achievements during high school[2]. The APS aims to inspire high school students to explore their career interests by creating an Academic Portfolio, which will be considered one of the criteria for college admission[4]. However, in practice, the pressure to excel in exams in Taiwan often leaves students with minimal time to explore and determine their desired field of study. With the goal of achieving the United Nations' Sustainable Development Goal 4 of Quality Education, our objective is to study the significant issues that high school students face in exploring college majors and career paths within the current educational environment, and to design a system that guarantees these students have access to inclusive, fair, and high-quality education that encourages lifelong learning. To accomplish this goal, we conducted an interview study which revealed that the students' primary obstacle was the absence of a platform that enables them to efficiently learn about and compare universities and departments. Additionally, they expressed a desire to have access to more information about college majors and the implications of their choices from the experiences shared by individuals who have gone through the same process.

Based on these findings, we applied a user-centered design process that included the development of personas and key design requirements, several rounds of idea generation and prototyping, and assessment based on these design requirements. Our proposed system is named *UniCompass*, and it comprises three main features:

College Matching, *Acquiring Experience*, and *E-Portfolio Planning*, which are intended to prevent high school students from going to multiple websites to search for information related to departments, saving time to integrate information, and assisting them in quickly discovering suitable college majors and career options to reduce the impact of the current educational environment where they may lack time to explore college majors. We introduce our design process and UniCompass in more detail below.

2 DESIGN PROCESS

2.1 User Research

We began by studying the difficulties high school students face when considering their future career paths and college majors. To gather information, we initially interviewed seven university students. Our goal was for them to share their personal experiences, needs, and obstacles encountered during their own exploration of potential college majors that they were interested in and suitable for. We specifically targeted students who had previously researched university department information before enrolling and still had clear memories of that process. We chose to interview university students instead of high school students in this phase of the study because they had already gone through the entire process of exploring college majors, from high school to just starting university. This allowed them to reflect on what could have been improved to make the process more effective and efficient, as well as comment on any discrepancies about their chosen major before and after entering university. The interviews were conducted using a semi-structured format and lasted between 45-60 minutes. We used affinity diagramming (Figure 1) to analyze the transcripts and identified several key themes. These results were then used to create personas and identify design requirements.

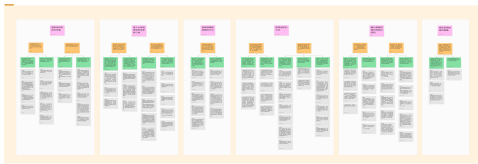


Figure 1: The affinity diagrams.

2.2 Summary of Key Findings

Our research yielded many findings, but we have chosen to focus on the two most critical ones for our design, which are presented below.

2.2.1 Lack of a Platform to Efficiently Learn about and Compare Universities and Departments. Participants indicated that when choosing a university and department, they primarily consider the learning threshold, career opportunities, competitiveness, salary level, and professional level of the department. When choosing a university, students take into account a variety of factors, including the university's facilities, human resources, learning resources, and location. These factors can have a major impact on students' career paths as well as their living situations, and as a result, their choices

of university and department. As P01 stated, *"If I want to learn marketing, most of the internship opportunities are in northern Taiwan, so I don't want to stay in the south."* However, currently, students face challenges in finding a user-friendly platform that consolidates information for comparing different universities, and thus the process of integrating information and comparison has been deemed laborious and troublesome, as P02 mentioned, *"Although each school's official website has the information, if I need to research 10 different departments, I would have to visit 10 different websites, which is quite cumbersome."* However, when they had not done this research beforehand, it resulted in students not fully understanding the department, which in turn caused difficulty in studying and a loss of interest and motivation. Since they have limited free time outside of their high-pressure and intensive school work, a platform that allows for efficient exploration of options is especially crucial.

2.2.2 Lack of Diversity and the Sharing of Experiences from Predecessors. Participants revealed that they spent most of their high school years studying and preparing for exams, due to a belief among teachers, parents, and cram schools that students should be focused on academics and the availability of resources in tutorial classes. As P01 explained, *"During my high school years, I felt that engaging in any activities other than studying would be a waste of study time as my parents emphasized the importance of studying at home instead of going out."* This has resulted in a lack of opportunity for them to interact with university seniors who could share their experiences of studying a specific department or the process of exploration. Many of them wished they had more opportunities to gain insights and learn about the experiences from university seniors who had already started studying the department they were interested in. As a result, their understanding of the department was incomplete and disjointed, leading to a discrepancy between participants' expectations and reality when they entered university and causing disappointment. As P03 mentioned, *"The aspects of campus life that are not as visible online, such as the academic aspects of a specific department, such as workload, are more commonly discussed. However, there are fewer people who share information about the emotional and mental experiences."*

2.3 Design Concept Ideation

We decided to focus on the two aforementioned findings because these painpoints were caused by the limited time and opportunity to explore career paths due to the high-pressure and exam-oriented educational environment. To investigate ways to address these issues, we developed personas based on the characteristics and goals of the participants, generated design requirements to aid them, and held ideation sessions based on the generated design requirements. In developing the personas, we created a behavioral spectrum based on students' considerations for choosing universities and departments, methods of obtaining information, and concerns and constraints. We matched participants on the spectrum and identified patterns among them, ultimately resulting in the generation of personas.

Based on the characteristics and patterns of our persona, we have defined two key requirements for the system.

- The system should help high school students efficiently compare the differences between various universities and departments and identify the more suitable one.

- The system should assist high school students in learning about universities and departments from a variety of personal experiences shared by current university students or former students who have researched the information.

We then performed sketching ideation based on the key design requirements. We proposed several design concepts for each key requirement, and for each design concept, we developed variations (Figure 2). By comparing and discussing these design concepts and variations, we created prototypes and obtained individual feedback to improve them in the next iteration.

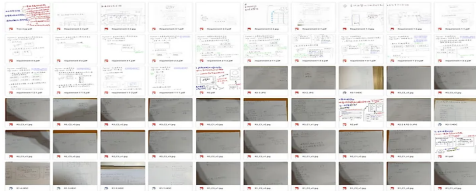


Figure 2: The sketches we drew during the ideation phase.

2.4 Prototype Modification and Additional Insight Discovery

We recruited a total of 11 participants to evaluate the prototype we created and obtain their feedback on the design concept, interaction flow, and usability. After each iteration, we modified the prototype based on the feedback we received. Through this process, in addition to enhancing the usability of the design, we also gained new insights that prompted us to revise our new design requirements. For example, a participant who was a high school student shared with us that the pressure of simultaneously preparing for both the Academic Portfolios and General Scholastic Ability Test (GSAT), which are both essential for college admission, is significant and that this had also affected their exploration of career paths: *"Preparing for the GSAT while also preparing for Academy Portfolio is not good for focusing on both at the same time."* (P13). For example, these students did not have enough time to think through how to prepare for Academic Portfolios and what to include. As P11 said, *"At the beginning, when I didn't know what to prepare or what to include, it was very frustrating, and I questioned why I had to do it."* As a result, without a clear understanding of what they wanted, they further mainly followed the blueprint that teachers or cram schools told them to follow to create the Academic Portfolios, which further reduced their incentive to explore on their own. This highlights the profound negative impact that Academic Portfolios have intensified on high school students' incentives and opportunities to spend time and effort exploring their interests and college majors. As a result, we added a new critical design requirement: **The system should provide advice and guidance on building Academic Portfolios**, in order to ease the burden of creating them, potentially creating more time for students to spend on exploring rather than studying.

Below, we present our proposed system, UniCompass.

3 UNICOMPASS

We chose the name UniCompass as the name combines "Uni" for university and "Compass" for guidance, signaling the tool's purpose of guiding students through the process of exploration to find the right university and major. UniCompass has three main components. (1) (Figure 3-a) which helps high school students efficiently discover the right major by comparing the differences between different universities and departments. (2) Acquiring Experience (Figure 3-b) which helps high school students to gain valuable insights from the personal experiences shared by current university students and experienced students/alumni, allowing them to engage in in-depth discussions about various college majors. (3) E-Portfolio Planning (Figure 3-c) offers high school students guidance and advice on creating Academic Portfolios, so that they have more time to efficiently explore various fields related to their chosen major and move towards their desired department.



Figure 3: Three main components of UniCompass: (a) College Matching, (b) Acquiring Experience, and (c) E-Portfolio Planning (Illustration by Storyset).

3.1 College Matching: Find the Right College Major and Compare Information

College Matching consists of three major features: *Search*, *Compare* and *Favorites* to help high school students explore college majors and efficiently search department information and compare. The Search component assists high school students in identifying an appropriate major if they do not have a clear preference. Specifically, it matches majors based on three dimensions: *GSAT Scores*, *Location Selection*, and *Aptitude Test*, which enable students to select which one to use. By collecting information on the 18 academic clusters and GSAT exam trends available online, and through AI technologies such as data analysis and machine learning, this system analyzes students' strengths, interests, and goals, which recommends the most suitable college majors to students based on the matching results page and explains why those majors are a good fit for them (Figure 4). High school students can also search for specific departments directly on UniCompass if they have a specific target in mind. In all of these cases, they can add the departments they are interested in to their Favorites for easy viewing later.

Given that high school students often struggle with organizing the information they collect from various sources online, UniCompass includes a *"Department Comparison Form"* (Figure 5). To make the exploration process easier for high school students, the Department Comparison Form has a user-friendly interface that enables students to efficiently side-by-side compare the characteristics of

different departments. The form enables users to tailor the information displayed to their customized needs and thus more effectively evaluate the departments of interest.

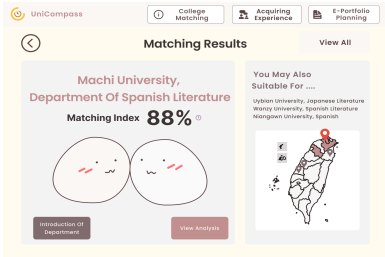


Figure 4: The matching results page indicates the college major that best matches the high school student’s interests.

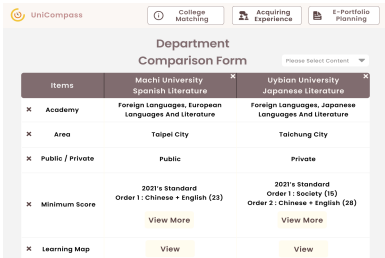


Figure 5: High school students can compare the differences between departments on Department Comparison Form directly and quickly.

3.2 Acquiring Experience: Obtain Information and Ask Questions

Given that high school students often struggle to find enough information about the departments they are interested in due to limited time and uneven distribution of information across departments, the Acquiring Experience component offers two sub-features, *Forum* and *Chatroom*, which allow students to learn about different departments and universities from the experiences of others. The Forum function (Figure 6) allows high school students to ask questions and receive subjective opinions from university students if they are unsure about a particular department. This can help them explore and clarify their desired department as they gather information. The Chatroom feature (Figure 7) allows high school students to establish contact with university students in the related departments and learn more about university life. It aims to solve more specific and customized issues not covered in the forum. Through previous iteration interviews, we have learned about that certain university students were helped by their university seniors when they were still in high school and their experience has inspired these university students to give back and provide assistance to current high school students. This is also one motivation for university students to provide information and assistance to high school students on UniCompass. A reward system has also been

set up so that university students can get rewards by responding to the inquiries of senior high school students. These rewards are based on the quality and timeliness of their answers. In addition, we will use a text filtering mechanism in the forum and chatroom functions to prohibit users from posting sensitive words. We also have a reporting mechanism to prevent users from making inappropriate comments or using the platform for purposes other than its original intention, in order to prevent it from becoming a dating app-like function.

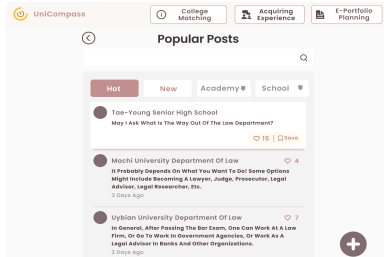


Figure 6: The Forum feature allows high school students to share information and ask questions.

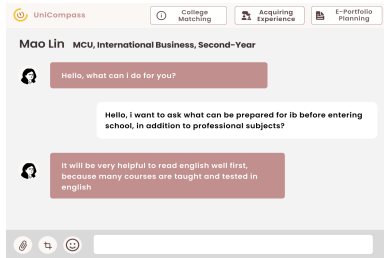


Figure 7: The Chatroom feature allows high school students to send messages to university students who are currently enrolled in the related department (Profile illustration by Open Peeps).

3.3 E-Portfolio Planning: Get Directions and Plan Schedules

E-Portfolio Planning contains *Suggestions* and *Schedule* (Figure 8) and tools to manage their to-do lists (Figure 9) to help high school students to create Academic Portfolios. Specifically, once high school students have set their goals on the Suggestions page, UniCompass will provide suggestions for the general direction of their Academic Portfolios. This addresses the problem of high school students who have a goal but are unsure of where to start creating their Academic Portfolios and helps them move forward with their exploration. The Schedule page helps high school students organize their tasks by creating personalized to-do lists and providing reminders for approaching deadlines. High school students often have limited time to explore departments of interest and may only begin preparing their Academic Portfolios shortly before the semester’s due date, resulting in portfolios that do not

effectively achieve their purpose. This feature is designed to assist high school students in efficiently planning their schedules, exploring university departments, and organizing their Academic Portfolios in a structured manner.

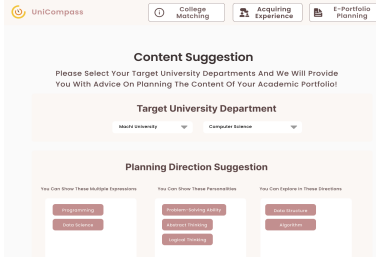


Figure 8: Advice for creating Academic Portfolio files on the Suggestions page provide directions that can be explored further.

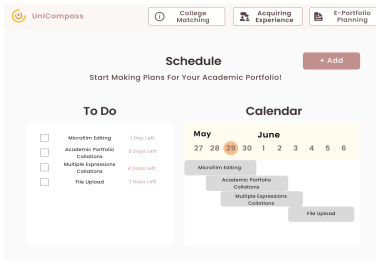


Figure 9: Schedule allows high school students to plan their schedules, explore university departments efficiently, and organize their Academic Programs in an orderly manner.

4 USER EVALUATION

We present results from the latest user evaluation of UniCompass, which is an ongoing process. Using the think-aloud technique, we recruited 11 high school and university students to evaluate the platform. Participants were asked to complete six tasks related to searching and exploring departments, assessing department suitability, obtaining and evaluating information using the Forum and Chatroom feature, and planning and scheduling for their academic portfolio.

All of the six tasks were completed successfully by the participants, with a 100% success rate, except for one task related to exploring departments, which had varying interpretations among participants. The system's overall usability, as measured by the SUS (System Usability Scale (SUS)[1]) score, was 78.9, indicating that it is easy to use. Participants provided positive feedback on UniCompass, stating they would use it regularly (or would have used it regularly if they were still high school students), found it user-friendly and well-integrated, and that it helped them explore university departments. For instance, the Search and Compare features in College Matching were particularly praised for their efficiency in finding and comparing departments, as P11 said, *"I used to search for only one university and department at a time, which was inconvenient. But*

with the College Matching function on the website, I can now avoid that and find what I need quickly."

Participants also appreciated the ability to directly learn from current university students, as they found their information more valuable and useful than that provided by official school websites. As P12 said, *"Current websites offer objective information, but there is no space for conversation with university students. But UniCompass has the Acquiring Experience function, which resolves it."* The participants also found the E-Portfolio Planning feature helpful, as it offers more practical and specific guidance. P16 noted, *"The information offered by the school is too general, but this Suggestion function provides me with more different perspectives."*

While participants had positive feedback on the platform, they also suggested areas for improvement, such as more detailed information on "tags" beyond academics and universities and more clarity in the mechanism for generating suggestions to increase trust in the platform's credibility.

5 CONCLUSION AND FUTURE WORK

In Taiwan, the high pressure to succeed academically leads students to prioritize majors with high earning potential and social prestige over their personal interests. This, along with the heavy emphasis on examination scores in college admissions, causes many students to attend cram schools, which limits their time to explore university departments and career paths. This limitation is further exacerbated by the implementation of the Academic Portfolio System, which puts extra pressure on students to excel in creating these portfolios, further reducing their time to explore the departments they are truly interested in. We conducted an interview study to better understand the challenges high school students face when trying to decide on a major for college in the short amount of time they have left and found that students most want the ability to quickly compare different options and gain insight from the experiences of others. We developed UniCompass through a series of iterative design phases; its three main features are College Matching, Acquiring Experience, and E-Portfolio Planning. Based on the results of our evaluation, UniCompass is well-received by its users, who find it helpful that the platform allows them to compare different departments side-by-side and gain insight into these departments from the perspective of presently enrolled students. In the future, it is anticipated that more research will be conducted on global educational concerns, resulting in more efficient methods for fostering student development. Additionally, Taiwan should work to find solutions to the education problem. In light of this, we feel that the value of UniCompass is founded in Sustainable Development Goal 4 of the United Nations, Quality Education. Our goal is to make UniCompass even more useful for Taiwanese high school students in the face of the increasingly competitive and test-centric curriculum in the country.

ACKNOWLEDGMENTS

We are grateful to Professor Yung-Ju Chang for his guidance and support throughout our project. We also appreciate the high school and university students who participated in our study, as their feedback helped improve our work. In addition, the utilization

of illustrations from Storyset and Open Peeps helped us make a visually appealing prototype.

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